Sports Resource Management System

Project Proposal for UCS503P
Submitted to: Ms. Anushka Chaudhary
Thapar Institute of Engineering and Technology

Soumya Smita Sahoo, AIML* Himasri Chadalavada, AIML†
Gaganpreet Kaur, AIML‡

September 2026

1 Higher-order goal *. . . efficient campus sports management*

This project aims to improve the accessibility, organization, and management of sports resources within a university campus. Sports facilities and equipment may be available, but students, coaches, and sports administration may face difficulty in knowing their availability, requesting equipment, booking facilities, tracking resources, and communicating requirements.

The immediate engineering objective is to translate these fragmented activities into a centralized and trackable mobile application that supports equipment management, facility booking, equipment requests, and resource tracking.

The application is designed for different categories of users including students or players, coaches, sports officers, administrators, and maintenance staff.

2 Time-to-value *. . . primary heuristic*

- We will deliver meaningful increments quickly by first implementing the core mobile workflows for login, equipment management, facility management, booking, and equipment requests.
- Development will be carried out incrementally so that the basic application and backend can be integrated before adding secondary modules.
- The initial prototype will focus on the most important equipment and facility management operations rather than attempting to implement every feature simultaneously.
- Additional functionality such as administrator management, maintenance tracking, feedback, and notifications will be added in subsequent iterations.
- Success in the initial stage is defined by having a working mobile prototype with reliable communication between the application, backend, and database.

*Roll No: 1024240041
†Roll No: 1024240038
‡Roll No: 1024240044

3 Problem Statement

Students and athletes often depend on sports offices or informal communication to access campus sports equipment and facilities. This can lead to uncertainty regarding availability, booking conflicts, difficulty in tracking equipment, and delays in handling resource requests.

Common pain points include:

- **Fragmentation:** Equipment requests, facility bookings, and resource-related communication may be handled through different channels such as physical registers, messages, calls, or informal communication.
- **Low availability transparency:** Students and coaches may not know whether a particular equipment item or facility is currently available.
- **Booking conflicts:** Multiple users may attempt to book the same facility for overlapping time periods, requiring manual coordination.
- **Poor equipment tracking:** Sports administration may face difficulty tracking equipment quantities, availability, condition, requests, and usage.
- **Equipment identification:** When equipment requires repair, maintenance staff need to identify the exact item. A unique Equipment ID can be used for this purpose.
- **Request management:** Equipment requests need to be stored and managed systematically so that sports staff can review and process them.
- **Limited centralized information:** Users, equipment, facilities, bookings, and requests should be connected through a centralized database.

4 Proposed Solution *. . . Sports Resource Management Mobile Application*

4.1 Overview

Build a cross-platform mobile application that provides role-based access to sports resource management functionality.

The proposed system contains the following components:

- **Student/Player App:** Students can log in, view sports equipment, check availability, request equipment, view facilities, and book available facilities.
- **Coach Module:** Coaches can view facilities, check availability, book facilities for sports activities, and request required equipment.
- **Sports Officer Module:** Sports officers can manage equipment and facilities, review equipment requests, and manage booking-related information.
- **Maintenance Module:** Maintenance staff will be able to identify equipment using its Equipment ID and manage repair-related information. This module is currently under development.
- **Administrator Module:** The administrator role is included for system-level management such as user and role management. This functionality is currently under development.
- **Feedback and Notification Modules:** Feedback and notifications are part of the planned system design. These modules are currently under development.

The system is being developed as a mobile application only. There is no separate web-based administrator dashboard in the current project scope.

4.2 Core workflow

Equipment workflow

1. The student or coach logs into the mobile application.
2. The user views the available sports equipment.
3. The application displays relevant equipment information, including availability and status.
4. The user selects the required equipment and submits a request.
5. The Sports Officer reviews the submitted request.
6. If the request is approved, the equipment can be issued to the user.
7. The equipment record can be updated with relevant issue and return information.
8. If the equipment requires repair, its unique Equipment ID can be used to associate it with a maintenance record.

Facility booking workflow

1. The student or coach selects a sports facility.
2. The application displays information about the selected facility and its availability.
3. The user selects the required date and time slot.
4. The system checks the existing booking information to determine whether the selected slot is available.
5. If the slot is available, the booking information is stored in the database.
6. The user can subsequently access the booking information through the application.

Maintenance workflow

The maintenance process begins when an equipment item is reported as requiring attention. Maintenance staff first inspect the equipment to understand the problem. If repair is required, the equipment is marked as being under repair while the maintenance work is carried out. After the repair is completed, the equipment is inspected again and its status is updated. If the equipment is found to be usable, it is marked as available for future use.

Each maintenance record is associated with the corresponding Equipment ID so that maintenance staff can identify the exact equipment under repair.

This module is part of the planned system and is not fully functional in the current prototype.

4.3 Operational constraints *. . . campus mobile setting*

- The application should use a lightweight and responsive mobile interface suitable for commonly available student smartphones.
- Core functions such as equipment viewing, booking, and equipment requests should require minimal steps.
- The application should provide clear loading and error states when communication with the backend is delayed or unavailable.
- The system should prevent conflicting facility bookings.

- Equipment marked as unavailable or under maintenance should not be treated as normally available.
- The system will prioritize workflow correctness and reliable resource management rather than advanced prediction algorithms.
- The backend and database will remain separate from the mobile interface so that additional modules can be integrated later.

5 Solution Approach *... engineering focus*

This is an engineering course project, so the primary focus is on mobile workflow design, role-based access, resource management, database integration, system correctness, and iterative development rather than novel algorithmic research.

5.1 Resource availability and conflict checking

Facility and equipment availability will be determined using stored resource information, booking status, equipment quantity, condition, and maintenance status.

The backend will validate relevant availability information before storing booking-related records. This is intended to reduce conflicting facility reservations.

5.2 Equipment lifecycle management

Equipment records will contain a unique Equipment ID along with information such as name, category, quantity, condition, and status.

The equipment lifecycle begins when an item is available for use. When a user receives the equipment, its status can be recorded as issued. After the equipment is returned, its condition can be checked. The equipment can then either be made available again or associated with a maintenance process if repair is required.

The use of Equipment IDs also allows individual pieces of equipment to be distinguished when multiple similar resources are present.

5.3 Role-based access

The application begins with role selection followed by the login workflow.

The selected role determines the functionality available to the user.

The main roles are:

- Student / Player
- Coach
- Sports Officer
- Administrator
- Maintenance Staff

5.4 Mobile application architecture

The system uses a mobile client-server architecture.

The Flutter and Dart mobile application provides the interface through which users interact with the system. Requests from the application are handled by the Node.js backend, which contains the server-side application logic. The backend communicates with PostgreSQL whenever data needs to be stored or retrieved.

This separation allows the mobile application, backend, and database to be developed and maintained as separate components.

5.5 Technology stack

The project uses the following technologies:

- **Flutter:** Cross-platform mobile application framework.
- **Dart:** Programming language used for the mobile application.
- **Node.js:** Backend runtime and server-side implementation.
- **PostgreSQL:** Relational database for persistent application data.
- **Visual Studio Code:** Development environment.
- **Git and GitHub:** Source-code version control and collaboration.

6 Evaluation Criterion *. . . measurable after implementation*

The current submission is a prototype-stage project. Therefore, formal evaluation results are not claimed at this stage.

After the remaining modules have been implemented, the system can be evaluated using functional and usability-oriented metrics.

6.1 Primary evaluation criteria

Core workflow correctness:

The main workflows should operate correctly from the mobile interface through the backend and database.

The following should be verified:

- Successful role selection and login.
- Correct display of equipment information.
- Correct submission and storage of equipment requests.
- Correct display of facilities.
- Correct facility booking.
- Correct handling of booking conflicts.
- Correct storage and retrieval of booking information.

6.2 Secondary evaluation criteria

- **Availability accuracy:** Equipment and facility availability shown by the application should correspond to the database records.
- **Booking correctness:** The system should not confirm conflicting bookings.
- **Request correctness:** Equipment requests should be stored with the correct user and request information.
- **Equipment identification:** Equipment IDs should uniquely identify equipment records.
- **Usability:** Common workflows such as viewing resources, making a booking, and submitting a request should be understandable to users.
- **Reliability:** Core application functionality should remain usable during normal prototype testing.

6.3 Future validation plan

Once the complete prototype is available, representative users can be asked to perform common tasks such as:

- Log in using their assigned role.
- Search or view equipment.
- Submit an equipment request.
- View facility availability.
- Book a facility.
- Attempt a conflicting booking.
- Review booking information.
- Test equipment maintenance identification using Equipment ID.

The results of such testing will be documented in a later project iteration.

7 Scalability *. . . with foresight*

The system should support growth in the number of students, coaches, equipment items, facilities, bookings, and requests.

1. **Scalable architecture:** The mobile client, backend, and database are separated so that each layer can be developed and optimized independently.
2. **Database scalability:** Database indexing can be introduced for frequently searched equipment, facility, and booking information.
3. **Large resource lists:** Pagination can be introduced when the equipment and booking catalogue becomes large.
4. **Efficient search:** Search and filtering can be improved for large campus inventories.
5. **Backend scalability:** The Node.js backend can be extended to support a larger number of concurrent mobile users.
6. **Future deployment:** The architecture can later be deployed using cloud-hosted backend and managed database services.

8 Engine availability heuristic

A mature set of development tools and technologies is available to implement the proposed mobile application:

- **Flutter** for cross-platform mobile development.
- **Dart** for application programming.
- **Node.js** for backend API development.
- **PostgreSQL** for structured relational data.
- **Git/GitHub** for version control and collaboration.
- Authentication mechanisms for role-based access.
- Testing tools for validating application and backend functionality.

These mature technologies allow the project to focus primarily on workflow design, resource management, database integration, and system correctness rather than developing new infrastructure.

9 Project scope and deliverables *. . . iteration-friendly*

9.1 Initial deliverable *. . . working prototype*

The initial prototype focuses on the following functionality:

- Flutter mobile application setup.
- Role selection.
- User login.
- Student / Player functionality.
- Coach functionality.
- Sports Officer functionality.
- Equipment catalogue and equipment information.
- Equipment availability.
- Equipment request submission.
- Facility listing.
- Facility availability.
- Basic facility booking.
- Node.js backend.
- PostgreSQL database integration.
- Backend and mobile application communication.

9.2 Subsequent deliverables

The following modules will be developed incrementally:

- Administrator functionality.
- Maintenance Staff interface.
- Equipment-specific maintenance tracking.
- Equipment damage and repair status tracking.
- Improved equipment issue and return tracking.
- Feedback and suggestion functionality.
- Notification functionality.
- Booking and equipment history.
- Improved search and filtering.
- Additional resource management features.
- Comprehensive testing and evaluation.

10 Risks and mitigations

- **Incorrect inventory information:** Maintain centralized equipment records and ensure that equipment status is updated through the application.
- **Double booking:** Perform availability validation before confirming a facility booking.
- **Equipment identification problems:** Assign a unique Equipment ID to each trackable equipment record.
- **Equipment condition disputes:** Maintain condition information when equipment is issued and returned. A more detailed condition workflow can be introduced later.
- **Incomplete prototype modules:** Prioritize core functionality first and implement administrator, maintenance, feedback, and notification modules incrementally.
- **Network/connectivity issues:** Provide appropriate loading, retry, timeout, and error states in the mobile application.
- **Device compatibility:** Test the mobile application on representative devices and screen sizes.
- **User adoption:** Keep common workflows such as login, resource viewing, booking, and equipment requests simple.

11 Summary

This project proposes a centralized mobile Sports Resource Management System for managing university sports equipment and facilities.

The application connects students, coaches, sports officers, administrators, and maintenance staff through role-based mobile workflows. Core functionality includes equipment management, facility availability, facility booking, and equipment requests.

The system is being developed using Flutter and Dart for the mobile application, Node.js for the backend, and PostgreSQL for the database. The separation between the mobile client, backend, and database provides a foundation for future expansion.

The current prototype focuses on the core working functionality. Administrator functionality, maintenance tracking, feedback, and notifications are planned as subsequent modules rather than being claimed as completed features.

The use of unique Equipment IDs will allow individual equipment items to be identified when they are issued, returned, damaged, or eventually sent for maintenance.

The project therefore focuses on practical software engineering requirements including multi-user workflows, role-based access, database design, mobile application development, backend integration, system correctness, iterative development, and future scalability.